

Efficient Bayesian Analysis of Longitudinal and Survival Data Using R-INLA

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Pre-Conference Course

2017-2020: Ph.D. in Public Health, Biostatistics, University of Bordeaux

Development of joint models for oncology.

2021-2024: Postdoctoral Fellow, KAUST University, Saudi Arabia

Biostatistical models using the INLA methodology.

2025-Now: Permanent Research Scientist at Inserm, University of Bordeaux

Expanding and applying the framework to clinical and epidemiology research.

Before we start: what to expect

- ▶ This course is built on recent research and materials published only months ago.
- ▶ New comes with trade-offs: you are getting state-of-the-art methods, but **not a course refined over decades of teaching**.
- ▶ **I am an active researcher, not an experienced lecturer**. I develop these methods and will share them as clearly as I can.
- ▶ **Your feedback is welcome**.
- ▶ Questions are never interruptions. **If something isn't clear, it is probably my fault, so ask**.

Why this course? 1/2

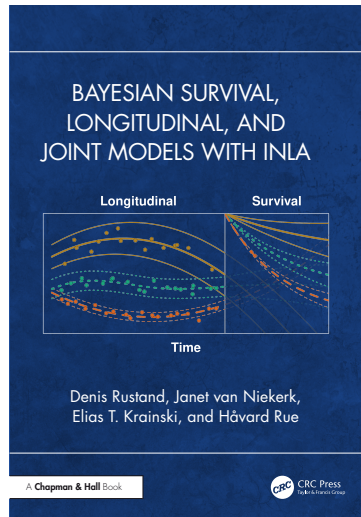
- ▶ **Context** Modern biomedical research where patients are followed over time.
- ▶ **Data** Repeated biomarker measurements and clinically relevant events.
- ▶ **Objective** We want to understand how the disease evolves, how treatments alter this trajectory and how markers and clinical events inform each other.
- ▶ **Tools** A unified framework to fit survival, longitudinal and joint models efficiently using **INLA** and **INLAjoint**.

Why this course? 2/2, the book

This course provides an overview of concepts and methodologies detailed in the new book:

“Bayesian survival, longitudinal and joint models with INLA”

- ▶ Chapman & Hall/CRC, May 2026
- ▶ An online free version is available at https://rustand.fr/INLA_book.



- ① **Event-time information** (death, progression, relapse, transplant or hospital readmission)
→ *terminal or recurrent, partially observed*
- ② **Longitudinal information** (repeated biomarkers, imaging summaries, lab values or symptoms)
→ hierarchical structure, *different distributions (continuous, counts, binary, ordinal)*
- ③ **Joint modeling of 1. and 2.**
→ *challenging computations, various forms*

→ INLA

Why not use MCMC for everything?

- ▶ Sampling-based Bayesian methods are general, but the computational burden can become substantial for rich joint models.
 - ▶ The models in this course are often high-dimensional and hierarchical, with many random effects, several likelihoods and rich dependence structures.
 - ▶ Recent INLA developments make these models much more accessible, with scalable computation for large cohorts and complex models.
- We can go beyond some practical limitations of alternative methods.

Computational advantage of INLA

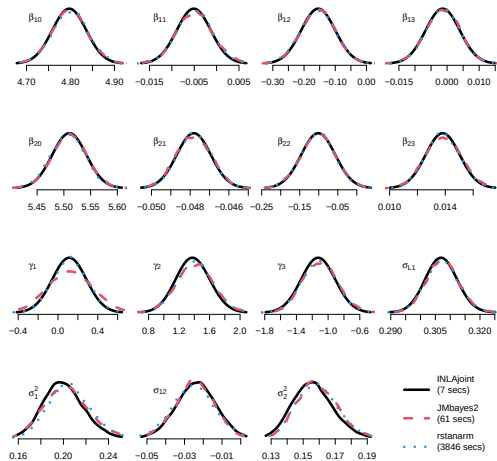
Computation time comparison (in seconds)

Model	INLA	MCMC (JAGS)	HMC (Stan)
Proportional hazards	0.6	15	67
Multi-state	0.8	155	94
Joint model	12	1754	949
Multivariate joint model	23	7105	9426

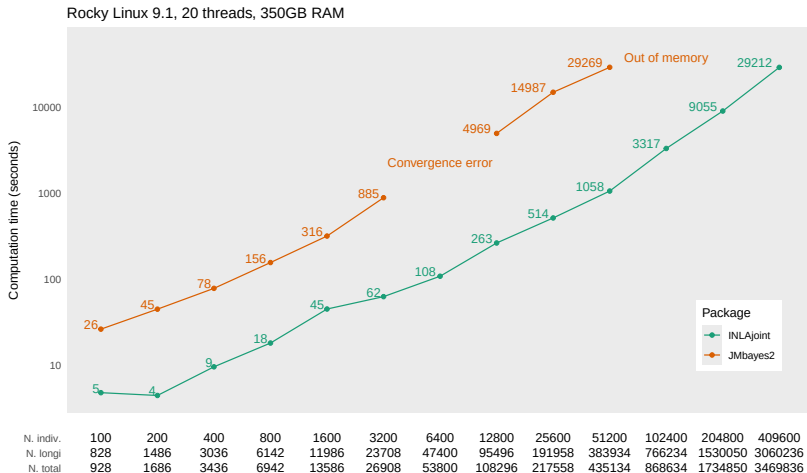
Alvares, D., Van Niekerk, J., Krainski, E. T., Rue, H., & Rustand, D. (2024). **Bayesian survival analysis with INLA**. *Statistics in medicine*, 43(20), 3975-4010.

Rustand, D., Van Niekerk, J., Krainski, E. T., Rue, H., & Proust-Lima, C. (2024). **Fast and flexible inference for joint models of multivariate longitudinal and survival data using integrated nested Laplace approximations**. *Biostatistics*, 25(2), 429-448.

Example: multivariate joint model



Comp. time scaling with data size



Bayesian inference and INLA

$$\pi(\boldsymbol{\theta} \mid \mathbf{D}) = \frac{\pi(\mathbf{D} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\pi(\mathbf{D})}$$

Component	Meaning in this course
$\pi(\boldsymbol{\theta} \mid \mathbf{D})$	Posterior distribution for fixed effects, random effects and hyperparameters
$\pi(\mathbf{D} \mid \boldsymbol{\theta})$	Likelihood from survival and longitudinal submodels
$\pi(\boldsymbol{\theta})$	Prior information and regularization
$\pi(\mathbf{D})$	Marginal likelihood, usually hard to compute exactly

INLA (Integrated Nested Laplace Approximations): a fast, accurate and deterministic alternative to MCMC for a broad class of models known as **latent Gaussian models**.

Instead of sampling, INLA directly calculates **highly accurate approximations** of the posterior marginals for each model parameter.

This approach is particularly efficient for the **hierarchical models** used in survival and longitudinal data analysis.

Rue H, Martino S, Chopin N. (2009). **Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations**. *Journal of the Royal Statistical Society: Series B* 71(2), 319–392.

Van Niekerk, J., Krainski, E., Rustand, D., and Rue, H. (2023). **A new avenue for Bayesian inference with INLA**. *Computational Statistics & Data Analysis* , 181, 107692.

Latent Gaussian model

The central idea is that a large family of useful models can be written in a common form.

$$\pi(\boldsymbol{\chi}, \boldsymbol{\theta} | \mathbf{D}) \propto \pi(\mathbf{D} | \boldsymbol{\chi}, \boldsymbol{\theta}) \pi(\boldsymbol{\chi} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta})$$

For outcome k on subject i at time t ,

$$g\{E[D_{ik}(t) \mid b_{ik}]\} = \eta_{ik}(t) = X_{ik}(t)^\top \beta_k + Z_{ik}(t)^\top b_{ik}.$$

This includes fixed effects, random effects, smooth functions and many familiar regression models.

INLA is designed for this latent Gaussian model family.

INLA stands for **Integrated Nested Laplace Approximations**.

- ① **Approximates** the posterior distribution of the **hyperparameters**.
- ② **Approximates** the conditional posteriors of the **latent effects**.
- ③ **Integrates** over the hyperparameters to recover **marginal posteriors**.

INLA **avoids** long **sampling-based computation** by exploiting the structure of latent Gaussian models and **sparse precision matrices**.

The R-INLA package

The R package INLA

The **R-INLA** package implements the **Integrated Nested Laplace Approximations** methodology in R. It computes **approximate posterior density functions** for latent Gaussian models.

The interface is **very general**. It offers robust capabilities for advanced models but it **requires specific knowledge and coding skills** to specify complex structures.

Installing R-INLA

```
install.packages(  
  "INLA",  
  repos = c(  
    getOption("repos"),  
    INLA = "https://inla.r-inla-download.org/R/stable"  
  ),  
  dep = TRUE  
)
```

The `inla()` function

The main function to fit a model is **`inla()`**. It requires three main arguments:

- ▶ **`formula`**,
- ▶ **`data`**,
- ▶ **`family`**.

Fixed effects are added with `+`. Random effects are specified using the **`f()`** function.

A logistic regression with a random intercept

```
library(INLA)
load("data/data_example.RData")
head(data_example)
  y      x1      x2 ID
1 0 -0.56047565 -0.99579872 1
2 1 -0.23017749 -1.03995504 1
3 1  1.55870831 -0.01798024 1
4 0  0.07050839 -0.13217513 1
5 1  0.12928774 -2.54934277 1
6 0  1.71506499  1.04057346 1

result_inla <- inla(
  y ~ x1 + x2 + f(ID, model = "iid"),
  data = data_example,
  family = "binomial"
)
```

Posterior summaries of the fixed effects

```
> summary(result_inla)
Time used:
  Pre = 0.278, Running = 0.287, Post = 0.0156, Total = 0.581
Fixed effects:
      mean      sd 0.025quant 0.5quant 0.975quant
(Intercept) -0.072 0.119      -0.307   -0.072      0.162
x1           1.096 0.096       0.909    1.095      1.286
x2          -1.109 0.096      -1.297   -1.109     -0.922

Model hyperparameters:
      mean      sd 0.025quant 0.5quant 0.975quant mode
Precision for ID 1.33 0.425      0.733    1.26      2.36 1.14
```

The summary shows the posterior **mean**, **sd** and credible intervals for each parameter. INLA uses precision instead of variances.

Extracting results

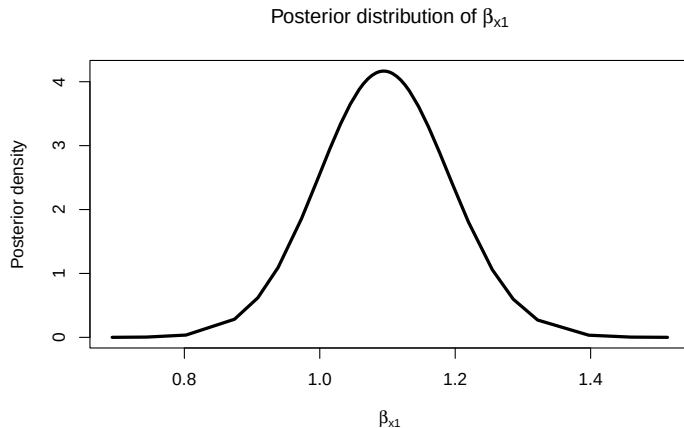
```
summary(result_inla)

result_inla$summary.fixed
result_inla$marginals.fixed

plot(result_inla$marginals.fixed$x1,
      type = "l")
```

For random effects or hyperparameters, replace **fixed** with **random**, **hyperpar**, **fitted.values** or **linear.predictor**.

Approximate marginal posterior density



Finding documentation internally

```
# Available likelihood families
names(inla.models()$likelihood)

# Documentation for a likelihood
inla.doc("zeroinflatedpoisson")

# Latent models for f()
names(inla.models()$latent)

# Available priors
names(inla.models()$prior)
```

The INLAjoint package

Why INLAjoint?

To address the **coding complexity of R-INLA** for multiple outcomes, the **INLAjoint** package was developed. It provides a user interface **tailored for survival, longitudinal and joint data analysis**.

It replaces **error-prone coding** with an **intuitive framework**. It also includes **post-processing functions** to extract predictions, hazard ratios and survival probabilities easily.

Installing INLAjoint

```
# CRAN version
install.packages("INLAjoint")

# Development version
devtools::install_github(
  "DenisRustand/INLAjoint"
)
```

One fitting function: `joint()`

In R-INLA, the main fitting function is **`inla()`**. In INLAjoint, it is **`joint()`**.

Use the components needed for the analysis:

- ▶ **`formSurv`** specifies survival models,
- ▶ **`formLong`** specifies longitudinal mixed models,
- ▶ **`assoc`** links longitudinal and survival components in a joint model.

The result is one fitted object for **`summary()`**, **`plot()`** and **`predict()`**. We will use the same workflow in each session.

Hyperparameter integration: `int.strategy`

The **`int.strategy`** option selects the integration strategy. Choices include **"auto"**, **"ccd"**, **"grid"** and **"eb"**.

```
joint(..., control = list(int.strategy = "eb"))
```

Empirical Bayes ("eb") simplifies the **last INLA step**: latent-effect marginals use the conditional posteriors at the hyperparameter mode, without averaging over hyperparameter uncertainty.

Hyperparameter uncertainty is still quantified. The hyperparameter posterior is approximated and can still be sampled.

Rustand et al. (2024) found **similar bias, variability and coverage** compared to the full integration, with reduced computation time.

Rustand, D., Van Niekerk, J., Krainski, E. T., Rue, H., & Proust-Lima, C. (2024). **Fast and flexible inference for joint models of multivariate longitudinal and survival data using integrated nested Laplace approximations.** *Biostatistics*, 25(2), 429-448.

From a fitted model to results

The `summary()` function

Here, **MODEL** denotes a fitted object returned by **joint()**.

```
summary(MODEL)
```

```
summary(MODEL, sdcor = TRUE, hr = TRUE)
```

Posterior summaries and credible intervals are grouped by model component.

- ▶ **sdcor = TRUE** displays standard deviations and correlations instead of variances and covariances.
- ▶ **hr = TRUE** displays survival fixed effects and association parameters on the hazard-ratio scale.

The plot() function

Plot the posterior marginals from the same fitted object:

```
plot(MODEL, priors = TRUE, sdcor = TRUE)
```

- ▶ **priors = TRUE** overlays prior densities on posterior marginals.
- ▶ **sdcor = TRUE** uses standard deviations and correlations for random-effect covariance parameters.

The returned **ggplot** objects can be customized. We will compare priors and posteriors in the model-specific sessions.

The predict() function

Obtain **patient-specific predictions**, integrating over random effects when present.

- ▶ **newData** supplies covariates and any observed marker histories.
- ▶ **horizon** sets how far ahead to predict.
- ▶ **survival = TRUE** adds survival probabilities to the hazard predictions.
- ▶ **inv.link = TRUE** returns longitudinal predictions on the response scale.

Updating the supplied marker history gives **dynamic predictions** in a joint model. We will illustrate this in the joint-model session.

Missing outcomes and missing covariates

Missing outcomes

A missing outcome can remain **NA**. It is treated as an **unobserved response** and can become a **prediction target** under the fitted model.

Missing covariates

The outcome model does not automatically define a distribution for a missing predictor. Covariates used in the linear predictor **must be observed or handled explicitly**, for example through imputation or a separate model.

- ▶ INLA provides [approximate Bayesian inference](#) for latent Gaussian models without posterior sampling.
- ▶ INLAjoint provides a specialized interface for [survival, longitudinal and joint models](#).
- ▶ The same fitted object provides [summaries, posterior plots and individual predictions](#).

Survival models with INLA

From the Cox model to INLA

$$\lambda_i(t) = \lambda_0(t) \exp\{X_i(t)^\top \beta\}$$

Cox regression estimates the **relative covariate effects** without specifying the baseline hazard $\lambda_0(t)$.

INLA instead works with a **full likelihood**, where the baseline hazard is estimated together with the covariate effects.

This is needed to get **absolute survival probabilities** and to fit more **complex survival models**.

To do this, we rewrite the survival likelihood using **events and observed time at risk**.

Cox DR. (1972). **Regression models and life-tables**. *Journal of the Royal Statistical Society: Series B* 34(2):187–220.

Follow-up as events and time at risk

Example: a patient dies after 2.5 years of follow-up.

Interval	Time at risk	Event
$(0, 1]$	1	0
$(1, 2]$	1	0
$(2, 3]$	0.5	1

This is the **counting-process view** of survival data: follow-up is decomposed into contributions of time at risk and events.

The three intervals are only for illustration, the same construction applies to any partition of follow-up.

The Poisson representation

For patient i in interval k , let D_{ik} be the **event count** and E_{ik} the **observed time at risk**.

Then $D_{ik} \sim \text{Poisson}(E_{ik}\lambda_{ik})$, so that $E(D_{ik}) = E_{ik}\lambda_{ik}$.

expected events = time at risk $\times$ hazard.

The hazard keeps the proportional-hazards structure:

$$\log \lambda_{ik} = \log \lambda_{0k} + X_{ik}^{\top} \beta.$$

The survival likelihood can therefore be fitted through a Poisson likelihood in INLA.

Holford TR. (1980). The analysis of rates and of survivorship using log-linear models. *Biometrics* 36(2):299–305.

Laird N, Olivier D. (1981). Covariance analysis of censored survival data using log-linear analysis techniques. *Journal of the American Statistical Association* 76(374):231–240.

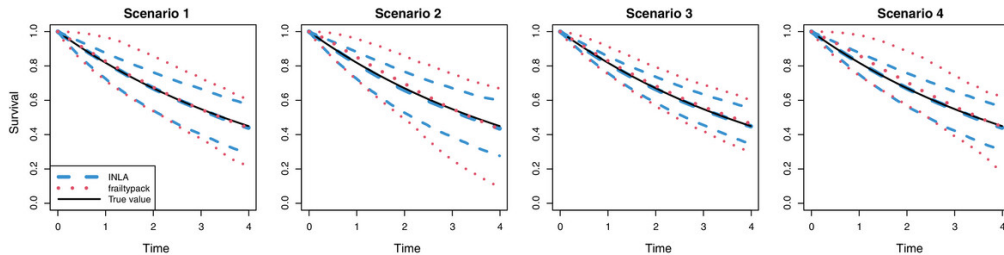
The key idea

Splitting follow-up into intervals gives a **Poisson representation of the survival model**.

- ▶ The **baseline hazard** can vary across intervals, with a fine time resolution and smoothness controlled by a random-walk prior.
- ▶ **Time-dependent covariates** can also vary across intervals and enter the hazard at their current value.

This is the bridge from proportional-hazards survival models to INLA.

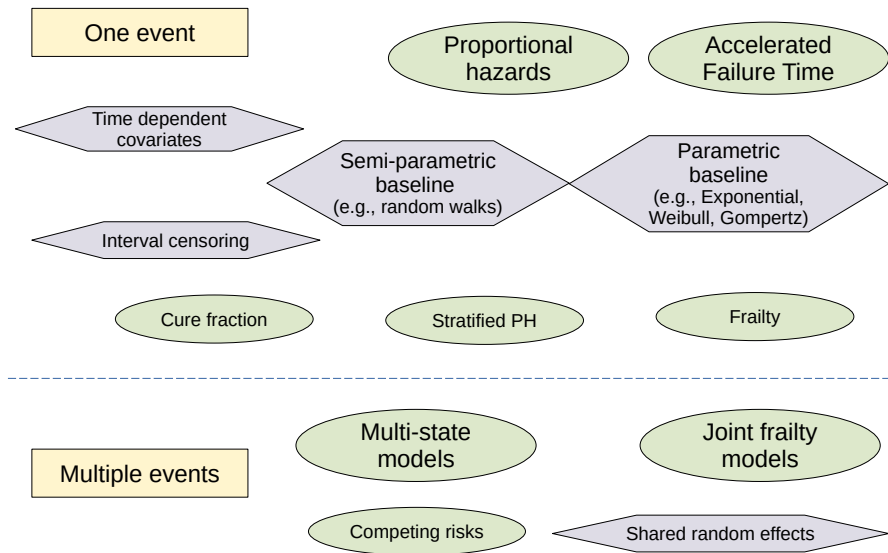
Baseline hazard comparison with splines



Rustand, D., van Niekerk, J., Rue, H., Tournigand, C., Rondeau, V., & Briollais, L. (2023). Bayesian estimation of two-part joint models for a longitudinal semicontinuous biomarker and a terminal event with INLA: Interests for cancer clinical trial evaluation. *Biometrical Journal*, 65(4), 2100322.

**Summary of survival and longitudinal models
available within the INLA/INLAjoint framework**

Survival Models for Time-To-Event Data



Longitudinal Models for Repeated Measurements

